Definition: The *simple random walk* on $\mathbb{Z}^d$ is given by a walker starting at $(0,\dots,0) \in \mathbb{Z}^d$, then at time t, moves from position

$$x(t) = (x_1,\dots,x_i,x_{i+1},\dots,x_d)$$

to position

$$x(t+1) = (x_1,\dots,x_i,x_{i+1},\dots,x_d)$$

with probability $1/(2d)$ for all $1 \le i \le d$ and every discrete time unit.

Our question is somewhat simple: when, if ever, does the walker return to the origin? This is the substance of Polya's Theorem and its generalization, Varopoulos's Theorem.

The first job is to formalize the setting for this question.

Markov Chains

Setup and Basic Definition

Definition: Let X be a finite or countable set. Let $P = (p(x,y))_{x,y\in X}$ be a *stochastic matrix*, a matrix satisfying $p(x,y) \ge 0$ for all $x,y \in X$ and

$$\sum_{y\in X} p(x,y) = 1$$

for all $x \in X$.

We call P the one-step transition probability matrix. The *initial distribution* is a map $\pi\colon X \to \mathbb{R}$ such that $\pi(x) \ge 0$ for all $x \in X$ and $\sum_{x\in X}\pi(x) = 1$. The triple (X,P,π) is called a *Markov chain*.

We may formalize this as follows. Let

$$\Omega = X^{\mathbb{N}}$$

be the space of trajectories. For a fixed $y_0,\dots,y_m \in X$, we let

$$C(y_0,\dots,y_m) = \bigcap_{j=0}^{m} Z_j^{-1}\left(\{y_j\}\right)$$

be the cylinder set with base tuple $(y_0,\dots,y_m) \in X^{m+1}$, where we let $Z_j\colon \Omega \to X$ be the projection map. Define a probability measure $\mathbb{P}_\pi$ by

$$\mathbb{P}_\pi(C(y_0,\dots,y_m)) = \pi(y_0)p(y_0,y_1)\cdots p(y_{m-1}y_m).$$

By the Kolmogorov Extension Theorem, we may uniquely extend $\mathbb{P}_\pi$ to the σ-algebra $\mathcal{B} \subseteq P(\Omega)$ generated by the cylinder sets. We will let $\mathbb{P}$ be this probability measure.

The projection maps Z_j are then X-valued random variables reflecting the position of the walker at time j. The projection maps have the special property known as memorylessness — that is,

$$\begin{aligned} p(x,y) &= \mathbb{P}(Z_{n+1} = y | Z_n = x) \\ &= \mathbb{P}(Z_{n+1} = y | Z_n = x \cap Z_{n-1} = x_{n-1} \cap \cdots \cap Z_0 = x_0). \end{aligned}$$

This is also known as the Markov property.

Conversely, it can be shown that if $(Z_n)_n$ is a sequence of X-valued random variables satisfying the Markov property, then there is a Markov chain (X,P,π) that can be found by taking

$$\pi(x) = \mathbb{P}(Z_0 = x)$$

and determining p as follows: if there exists $n \in \mathbb{N}$ such that $P(Z_n = x) > 0$, set $n(x)$ to be the minimal such value, and else set $n(x) = \infty$; then, set

$$p(x,y) = \begin{cases} P\Big(Z_{n(x)+1} = y | P\Big(Z_{n(x)=x}\Big)\Big) & n(x) < \infty \\ \Phi(y) & \text{else} \end{cases}$$

for all $x, y \in X$, where $\Phi \colon X \to [0,1]$ is a map such that $\sum_{y \in X} \Phi(y) = 1$. If $n(x) < \infty$ for all $x \in X$, then this Markov chain is unique.

Note that Φ is, in a sense, an atomic probability measure.

Definition: Given a Markov chain (X, P), we let $\mathcal{G} = \mathcal{G}(X,P) = (X,E)$ be the directed graph associated with (X,P) defined as follows. The vertex set is the state space of the Markov chain, and $E \subseteq X \times X$ is the set of pairs $(x,y) \in X \times X$ such that $p(x,y) \neq 0$.

For any $n \in \mathbb{N}$, we set $P^0 = I = \left(\delta_{x,y}\right)_{x,y \in X}$, and letting

$$\begin{aligned} P^n &= PP^{n-1} \\ &= \left(p^{(n)}(x,y)\right)_{x,y \in X} \end{aligned}$$

be the nth power of the transition matrix P. Notice that $p^{(1)}(x,y) = p(x,y)$, and

$$p^{(n+1)}(x,y) = \sum_{u \in X} p(x,u) p^{(n)}(u,y).$$

Provided $\pi(x) > 0$, one has that $p^{(n)}(x,y) = \mathbb{P}(Z_n = y | Z_0 = x)$. That is, $p^{(n)}(x,y)$ gives the probability that starting from state x one arrives in state y in n random steps. In particular, we have that $\mathbb{P}(Z_{k+n} = y | Z_k = x) = p^{(n)}(x,y)$.

Suppose that $\pi = \delta_{x_0}$ is the initial configuration, specifically expressing the point mass at the starting point. Let $v_n^x(\omega) = \delta_x(Z_n(\omega))$. Then, v_n^x equals 1 if at time n, the random walker with trajectory ω is exactly at x, and 0 otherwise. Then,

$$\begin{aligned} E[v_n^x] &= \int v_n^x(\omega)\, d\mathbb{P}_{x_0}(\omega) \\ &= \mathbb{P}_{x_0}\Big(\bigcup \{C(x_0, y_1, \ldots, y_{n-1}, x) : y_1, \ldots, y_{n-1} \in X\}\Big) \\ &= \sum_{y_1, \ldots, y_{n-1} \in X} \mathbb{P}_{x_0}(C(x_0, y_1, \ldots, y_{n-1}, x)) \\ &= \sum_{y_1, \ldots, y_{n-1} \in X} p(x_0, y_1) p(y_1, y_2) \cdots p(y_{n-1}, x) \qquad \text{(Definition of } \mathbb{P}_{x_0}.) \\ &= p^{(n)}(x_0, x). \end{aligned}$$

Thus, we have that

$$v^x = \sum_{n=0}^{\infty} v_n^x$$

represents the total number of times the random walker starting at x_0 visits x. The corresponding expected value is given by

$$E[v^x] = \sum_{n=0}^{\infty} p^{(n)}(x_0, x).$$

Irreducibility

Definition: A Markov chain (X, P, π) is called *irreducible* if for every $x, y \in X$, there exists some $n \in \mathbb{N} \setminus \{0\}$ such that $p^{(n)}(x, y) > 0$.

Equivalently, the associated graph for the Markov chain is connected. We call the power series

$$G(x, y|z) = \sum_{n=0}^{\infty} p^{(n)}(x, y) z^n$$

the *Green's function* for the Markov chain.

Proposition: Suppose (X, P, π) is irreducible, and $0 < t < \infty$. Then, either $G(x, y|t)$ converges for every $x, y \in X$ or diverges for every $x, y \in X$.

Proof. Let $x, y, x', y' \in X$, and suppose that $G(x, y|t) < \infty$. We will show that $G(x', y'|t) < \infty$.

By irreducibility, we may find $h, k \in \mathbb{N} \setminus \{0\}$ such that $p^{(h)}(x, x') > 0$ and $p^{(k)}(y', y) > 0$. Furthermore,

$$p^{(h+n+k)}(x, y) \geq p^{(h)}(x, x') p^{(n)}(x', y') p^{(k)}(y', y)$$

for all $n \in \mathbb{N}$. Thus,

$$\begin{aligned}
G(x, y|t) &= \sum_{m=0}^{\infty} p^{(m)}(x, y) t^m \\
&\geq \sum_{m=h+k}^{\infty} p^{(m)}(x, y) t^m \\
&= \sum_{n=0}^{\infty} p^{(h+k+n)}(x, y) t^{h+k+n} \\
&\geq \sum_{n=0}^{\infty} t^{h+k} t^n p^{(h)}(x, x') p^{(n)}(x', y') p^{(k)}(y', y) \\
&= p^{(h)}(x, x') p^{(k)}(y', y) t^{h+k} \sum_{n=0}^{\infty} p^{(n)}(x', y') t^n \\
&= p^{(h)}(x, x') p^{(k)}(y', y) t^{h+k} G(x', y'|t),
\end{aligned}$$

whence $G(x', y'|t) < \infty$. □

We will always assume that our Markov chains are irreducible. Set

$$\begin{aligned}
G(x, y) &= G(x, y|1) \\
&= \sum_{n=0}^{\infty} p^{(n)}(x, y) \\
&\in [0, \infty].
\end{aligned}$$

Given $x, y \in X$, the real random variable

$$T^x = \inf\{n \geq 1 : Z_n = x\}$$

is called a hitting time. This is the first time that the random walker with trajectory ω visits the state x. The quantity

$$f^{(n)}(x, y) = \mathbb{P}_x(T^y = n)$$

gives the probability that the random walker, after visiting x, arrives in y for the first time in n steps. This is called the hitting probability. Notice that $f^{(0)}(x, y) = 0$ for all $x, y \in X$. Denote by

$$F(x, y|z) = \sum_{n=1}^{\infty} f^{(n)}(x, y) z^n$$

as the associated generating function.

Proposition: Suppose the Markov chain (X, P, π) is irreducible. Let $x, y \in X$ be distinct elements. Then, the following hold:

(i) $G(x,y|z) = G(y,y|z)F(x,y|z)$;

(ii) $G(x,x|z) = 1 + G(x,x|z)F(x,x|z)$;

(iii) if $0 \leq t < \infty$ is such that $G(x,x|t) < \infty$ and $F(x,x|t) < 1$, then

$$G(x,x|t) = \frac{1}{1 - F(x,x|t)}.$$

Proof. Let $n \geq 1$, $Z_0 = x$, and suppose $Z_n = y$. Denote by k the value such that $T^y = k$. We have the equality

$$\{T^y = k\} = \{Z_k = y\} \cap \Big\{Z_j \neq y \text{ for all } j = 1, 2, \ldots, k-1\Big\}.$$

Using the Markov property, we may write

$$\begin{aligned}
p^{(n)}(x,y) &= \mathbb{P}_x(Z_n = y) \\
&= \sum_{k=1}^{n} \mathbb{P}_x(Z_n = y \cap T^y = k) \\
&= \sum_{k=1}^{n} \mathbb{P}_x(Z_n = y | T^y = k)\mathbb{P}_x(T^y = k) \\
&= \sum_{k=1}^{n} \mathbb{P}_x\Big(Z_n = y | Z_k = y \cap Z_j \neq y \text{ for all } j = 1, \ldots, k-1\Big)\mathbb{P}_x(T^y = k) \\
&= \sum_{k=1}^{n} \mathbb{P}_x(Z_n = y | Z_k = y)\mathbb{P}_x(T^y = k) \\
&= \sum_{k=1}^{n} p^{(n-k)}(y,y) f^{(k)}(x,y).
\end{aligned}$$

Thus, we have

$$p^{(n)}(x,y) = \sum_{k=0}^{n} p^{(n-k)}(y,y) f^{(k)}(x,y),$$

so that

$$\begin{aligned}
G(x,y|z) &= \sum_{n=0}^{\infty} p^{(n)}(x,y) z^n \\
&= p^{(0)}(x,y) + \sum_{n=1}^{\infty} p^{(n)}(x,y) z^n \\
&= \delta_x(y) + \sum_{n=1}^{\infty} \sum_{k=0}^{n} p^{(n-k)}(y,y) f^{(k)}(x,y) z^n \\
&= \delta_x(y) + G(y,y|z)F(x,y|z).
\end{aligned}$$

Now, if $0 \leq t < \infty$ is such that $G(x,x|t) < \infty$ and $F(x,x|t) < 1$, then by a theorem of Mertens, we have that the series $G(x,x|t)F(x,x|t)$ converges, so that we get our desired result upon dividing. □

Recurrence and Transience

Definition: Let (X, P, π) be a Markov chain. A state $x \in X$ is said to be *recurrent* if the probability that the Markov chain returns to x is 1. Else, we say it is transient.

In symbols, x is recurrent if and only if

$$\mathbb{P}_x(Z_n = x \text{ for some } n \in \mathbb{N}) = \mathbb{P}_x\left(\bigcup_{n\geq 1}\{Z_n = x\}\right) = 1.$$

Equivalently, x is recurrent if and only if $P(T^x < \infty) = 1$.

For any $x, y \in X$, let

$$\begin{aligned} F(x,y) &= \mathbb{P}_x(Z_n = y \text{ for some } n \in \mathbb{N}) \\ &= \mathbb{P}_x\left(\bigcup_{n\geq 1}\{Z_n = y\}\right) \\ H(x,y) &= \mathbb{P}_x(Z_n = y \text{ infinitely often}) \\ &= \mathbb{P}_x\left(\limsup_{n\to\infty}\{Z_n = y\}\right). \end{aligned}$$

A state $x \in X$ is recurrent if and only if $F(x,x) = 1$. Notice that

$$\begin{aligned} F(x,y) &= \mathbb{P}_x\left(\bigcup_{n\geq 1}\{Z_n = y\}\right) \\ &= \sum_{n=1}^{\infty} \mathbb{P}_x(T^y = n) \\ &= \sum_{n=0}^{\infty} f^{(n)}(x,y) \\ &= F(x,y|1). \end{aligned}$$

Theorem (Zero-One Law for Markov Chains)**:** Let (X, P, π) be an irreducible Markov chain. For any $x \in X$, the following are equivalent:

(a) x is recurrent (that is, $F(x,x) = 1$);

(b) $G(x,x) = \infty$;

(c) $H(x,x) = 1$;

(d) $H(x,x) > 0$.

Proof. We recall the definition of

$$\begin{aligned} G(x,x) &= \sum_{n=0}^{\infty} p^{(n)}(x,x) \\ &= \sum_{n=0}^{\infty} \mathbb{P}_x(Z_n = x). \end{aligned}$$

By the Borel–Cantelli laws and independence of $(Z_n)_n$, we have that (b) implies (c), (c) implies (d), and (d) implies (b).

Now, we will show that (a) and (b) are equivalent. First, from Abel's theorem, we have that $F(x,x) = \lim_{t\nearrow 1} F(x,x|t)$.

If $G(x,x) < \infty$, then by Abel's theorem yet again, we have that $\lim_{t \nearrow 1} G(x,x|t)$, so that

$$\begin{aligned}\lim_{t \nearrow 1} \frac{1}{1 - F(x,x|t)} &= G(x,x) \\ &< \infty,\end{aligned}$$

so that $F(x,x) < 1$. This gives that (a) implies (b). Conversely, if $F(x,x) < 1$, and R is the radius of convergence of $G(x,x|z)$, then by Pringsheim's theorem, we have that $\lim_{t \nearrow R} G(x,x|t) = \infty$, so that

$$\begin{aligned}\lim_{t \nearrow R} G(x,x|t) &= \lim_{t \nearrow R} \frac{1}{1 - F(x,x|t)} \\ &\leq \frac{1}{1 - F(x,x)} \\ &< \infty,\end{aligned}$$

which is a contradiction. Thus, $R > 1$, so that $G(x,x) = G(x,x|1) < \infty$. □

Random Walks on Groups

Throughout, we will let G be a finitely generated group and $Y \subseteq G$ a finite symmetric generating set.

Basic Definitions

Definition: Let $(G,P) = (G,P,\pi)$ an irreducible Markov chain on G. We say (G,P) is a random walk on G provided that the transition probabilities satisfy the following conditions:

(i) $p(g,h) = p(h,g)$ for all $g,h \in G$;

(ii) $p(g'g, g'h) = p(g,h)$ for all $g,g',h \in G$;

(iii) $\sum_{h \in G} d_Y(g,h)^2 p(g,h) < \infty$ for all $g \in G$.

A random walk on (G,P,π) that satisfies $p(g,h) = 0$ whenever $g^{-1}h \notin Y$ is called a nearest neighbor walk.

Example: The simple random walk on G is the nearest neighbor random walk $(G,Q) = (G,Q,\delta_e)$ such that

$$q(g,h) = \begin{cases} \frac{1}{|Y|} & g^{-1}h \in Y \\ 0 & \text{else.} \end{cases}$$

We may associate an atomic probability measure μ on G to a random walk (G,P,π) by taking

$$\mu(g) = p(e,g)$$

for all $g \in G$. Notice that this satisfies

(i) $\mu(g) = \mu\left(g^{-1}\right)$ for all $g \in G$;

(ii) $\operatorname{supp}(\mu)$ generates G;

(iii) $\sum_{h \in G} d_Y(e,g)^2 \mu(g) = \sum_{h \in G} \ell_Y(g)^2 \mu(g) < \infty$ for all $g \in G$.

Moreover, (ii) can be reformulated as saying that

$$G = \bigcup_{n=0}^{\infty} \operatorname{supp}(\mu^{*n}),$$

where μ^{*n} denotes the nth convolution of μ, defined by setting

$$\mu^{*n}(g) = \sum_{h \in G} \mu^{*(n-1)}\left(gh^{-1}\right)\mu(h)$$

$$= \sum_{h \in G} \mu^{*(n-1)}(h)\mu\left(h^{-1}g\right).$$

We notice that $p^{(n)}(e,g) = \mu^{(n)}(g)$.

Recurrence and Transience of Simple Random Walks on $\mathbb{Z}^d$

Proposition: The simple random walks on $\mathbb{Z}$ and $\mathbb{Z}^2$ are recurrent.

Proof. A finite random path in $\mathbb{Z}$ is a sequence $\pi = (x_0, \ldots, x_n)$ with $x_0 = 0$ and $|x_{i+1} - x_i| = 1$ for all i. If $x_n = 0$, then we call the path closed. The number of paths is 2^n, and the number of closed paths is given by $\binom{2m}{m}$.

In particular, the probability $p_n = q^{(n)}(0,0)$ of returning to the origin after n steps is given by

$$p_n = \begin{cases} \frac{1}{2^{2m}}\binom{2m}{m} & n = 2m \\ 0 & \text{else.} \end{cases}$$

By Stirling's approximation, we have that

$$\begin{aligned} p_{2m} &\sim \frac{1}{2^{2m}} \frac{(2m)^{2m} e^{-2m} \sqrt{4\pi m}}{m^{2m} e^{-2m} 2\pi m} \\ &= \frac{C}{\sqrt{m}} \end{aligned}$$

for some constant C. Thus, in particular, we have

$$\begin{aligned} G(0,0) &= \sum_{n=0}^{\infty} p_n \\ &= \sum_{m=0}^{\infty} p_{2m} \\ &\sim \sum_{m=0}^{\infty} \frac{C}{\sqrt{m}} \\ &= \infty, \end{aligned}$$

so that 0 is recurrent. This gives the case of $\mathbb{Z}$.

For the case of $\mathbb{Z}^2$, we note that if we start from the origin, then the number of closed paths of length m is given by

$$\begin{aligned} \sum_{k=0}^{m} \frac{(2m)!}{k!k!(m-k)!(m-k)!} &= \sum_{k=0}^{m} \frac{(2m)!}{m!m!} \frac{m!m!}{k!k!(m-k)!(m-k)!} \\ &= \sum_{k=0}^{m} \binom{2m}{m}\binom{m}{k}^2 \\ &= \binom{2m}{m}^2. \end{aligned}$$

Thus, the probability p_n' of $q^{(n)}((0,0),(0,0))$ returning to the origin after n steps is

$$p_n' = \begin{cases} \left(\frac{1}{2^{2n}}\binom{2m}{m}\right)^2 & n = 2m \\ 0 & \text{else.} \end{cases}$$

Thus, $p'_n \sim \frac{1}{m}$, and

$$\begin{aligned} G((0,0),(0,0)) &= \sum_{m=0}^{\infty} p'_{2m} \\ &= \infty. \end{aligned}$$

□

This pattern stops at $\mathbb{Z}^2$, as we will display now.

Theorem: The simple random walk on $\mathbb{Z}^3$ is transient.

Proof. We write

$$\begin{aligned} p''_{2m} &= q^{(2m)}((0,0,0),(0,0,0)) \\ &= \frac{1}{6^{2m}} \sum_{\substack{j,k\geq 0 \\ j+k\leq m}} \frac{(2m)!}{j!j!k!k!(m-j-k)!(m-j-k)!} \\ &= \frac{1}{2^{2m}}\binom{2m}{m} \sum_{\substack{j,k\geq 0 \\ j+k\leq m}} \frac{1}{3^{2m}}\left(\frac{m!}{j!k!(m-j-k)!}\right)^2. \end{aligned}$$

Since $a!b! \geq (a+1)!(b-1)!$, we have that for all $j,k \geq 0$ with $j+k \leq m$ that

$$\frac{m!}{j!k!(m-j-k)!} \leq \frac{m!}{\left\lfloor \frac{m}{3} \right\rfloor! \left\lfloor \frac{m}{3} \right\rfloor! \left\lfloor \frac{m}{3} \right\rfloor!}.$$

Thus, since

$$\sum_{\substack{j,k\geq 0 \\ j+k\leq m}} \frac{m!}{j!k!(m-j-k)!} = 3^m,$$

it follows (upon more tedious computation) that

$$p''_{2m} \leq \frac{1}{2^{2m}3^m} \frac{(2m)!}{m!\left(\left\lfloor \frac{m}{3} \right\rfloor\right)^3}.$$

Once again applying Stirling's approximation gives that

$$p''_{2m} \preceq \frac{C}{m^{3/2}},$$

where $a_k \preceq b_k$ when $\lim_{k\to\infty} \frac{a_k}{b_k} \leq 1$. Thus, we have

$$\begin{aligned} G((0,0,0),(0,0,0)) &= \sum_{m=0}^{\infty} p''_{2m} \\ &< \infty, \end{aligned}$$

so the random walk is transient. □

Proof Strategy for the Varopoulos Theorem

Our goal is to prove the Varopoulos Theorem, which says the following.

Theorem (Varopoulos)**:** Let G be a finitely generated group, and suppose the simple random walk on G with respect to some finite symmetric generating set is recurrent. Then, either

(i) G is finite;

(ii) G is virtually $\mathbb{Z}$;

(iii) G is virtually $\mathbb{Z}^2$.

The proof strategy will be as follows. We will develop the notion of the network associated to the Markov chain on G, and a special recurrence criterion to show that the simple random walk on any group whose growth is at most quadratic is recurrent. Meanwhile if the growth of G is at least cubic, we will explicitly construct a transient (not necessarily simple) random walk on G. Upon showing that the recurrence of a generic random walk on G is equivalent to the recurrence of the simple random walk, this will allow us to use Gromov's Theorem and the Bass–Guivarch formula to determine that the unique groups whose growth is at most quadratic are either finite, virtually $\mathbb{Z}$, or virtually $\mathbb{Z}^2$.

Recurrence Criteria

Network of a Markov Chain

Definition: We say a Markov chain (X, P, π) is reversible if there exists $m\colon X \to (0, \infty)$ such that

$$m(x)p(x,y) = m(y)p(y,x)$$

for all $x, y \in X$. The quantity

$$a(x,y) = m(x)p(x,y)$$

is called the conductance between x and y.

Example: If G is a finitely generated group, Y a finite symmetric generating subset, and (G, Q) the simple random walk on G associated to Y, then $q(g,h) = q(h,g)$ for all $g, h \in G$, so $m(g) = 1$ for all $g \in G$ defines an invariant measure.

If (X, E) is the directed graph associated with a reversible Markov chain (X, P), then we may in fact consider it as an *undirected* graph whose edges are the elements $\{x, y\}$ with $p(x,y) \neq 0$. Then,

$$\begin{aligned}\sum_{y\in X} a(x,y) &= \sum_{y\in X} m(x)p(x,y)\\ &= m(x)\sum_{y\in X} p(x,y)\\ &= m(x),\end{aligned}$$

so that

$$\begin{aligned}p(x,y) &= \frac{a(y,x)}{m(x)}\\ &= \frac{a(y,x)}{\sum_{z\in X} a(x,z)}.\end{aligned}$$

In particular, the transition matrix of any reversible Markov chain is uniquely determined by its conductances. Given an undirected graph with reversible Markov chain, we will fix an orientation on the edge set, and write $e = \{e^+, e^-\}$, where e^+ is the initial vertex and e^- is the terminal vertex.

Definition: The function

$$r(e) = \frac{1}{a(e^+, e^-)}$$

is called the resistance. The triple $\mathcal{N} = (X, E, r)$ is called a network.

References

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